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Theoretical Physics

Complex Networks

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Introduction

1 | Definitions

In this chapter, we will cover the basic definitions and concepts related to complex networks. We will start with fundamental graph concepts, then explore various network representations, discuss graph isomorphism and permutation, and finally delve into connectedness, distances, subgraphs, planarity, homomorphism, network reduction, and spectral analysis of networks.

1.1 | Basic Graph Concepts

Definition 1.1 (Graph). A graph $G(V, L)$ is a finite nonempty set V of p points (vertexes or nodes) and a set of k couples of distinct points L (links, lines, or edges).

Remark (General Rules).

- *In general, networks have no loops (self-connections).*
- *There are no multigraphs, meaning no multiple links between the same nodes.*
- *We can, however, consider multiple links as weights or as multiple layers (multiplex).*

1.1.1 | Directed and Undirected Graphs

Directed graphs (DiGraphs) are graphs with directed edges, known as arcs, indicating a “from-to” relationship. They are used to describe hierarchical relationships, fluxes, or the passing of messages and goods. Undirected graphs contain only symmetric links and are used to describe symmetrical relations such as friendship, chemical bonds, or statistical correlation. Note that directed graphs can be symmetrized; while this loses some information, it allows the use of simpler properties and algorithms.

1.1.2 | Walk, Trail, Path, and Cycle

Let us consider a graph with nodes i and j . We can define different types of connections between these nodes:

- **Walk:** A node and edge sequence between two nodes.
- **Trail:** A walk with all distinct links.
- **Path:** A walk with all distinct nodes and links.
- **Cycle:** A closed path with $N_c \geq 3$ nodes.

This classification is important for understanding the structure of networks and for defining concepts such as connectedness and distances. Different algorithms are used to find walks, trails, paths, and cycles in a graph, which can be crucial for applications like network traversal, routing, and community detection.

1.2 | Network Representations

It is important to note that a graph can be represented in various ways, each with its own advantages and disadvantages. The choice of representation often depends on the specific application and the properties of the graph being studied.

1.2.1 | Adjacency Matrix

The most common representation of a graph is the adjacency matrix, which is a square matrix that encodes the presence or absence of links between nodes. For unweighted graphs, the entries are binary (0 or 1), while for weighted graphs, the entries can take on a range of values corresponding to the weights of the links.

Definition 1.2 (Adjacency Matrix). The adjacency matrix A is an $N \times N$ matrix representing N nodes. For unweighted networks, $A_{ij} = 1$ indicates a topological link from node i to node j , with $A_{ii} = 0$ to indicate no loops. For weighted networks, $A_{ij} = w_{ij}$ where w_{ij} is the weight of the link.

Directed Graph

Consider the following directed graph with 6 nodes, we can represent it with the following adjacency matrix:

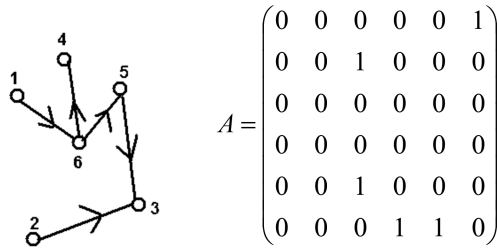


Figure 1.1: A directed graph and its corresponding non-symmetric adjacency matrix.

Undirected Graph

Consider the following undirected graph with 6 nodes, we can represent it with the following adjacency matrix:

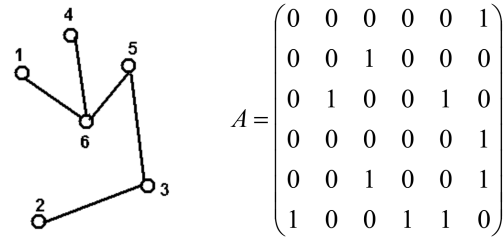


Figure 1.2: An undirected graph and its corresponding symmetric adjacency matrix.

Remark (Logical Operators on Adjacency Matrices). We can apply element-wise logical operators on adjacency matrices:

- Network symmetrization uses a logical OR: $A' = \max(A, A^T) = A \cup A^T$.
- Identifying reciprocal links uses a logical AND: $A'' = \min(A, A^T) = A \cap A^T$.
- Identifying strictly directed links uses logical XOR: $XOR(A, A^T)$.

1.2.2 | Weighted Network

In a weighted network, the adjacency matrix contains values that represent the strength of the connections between nodes:

$$A_{ij} = w_{ij} \text{ link from node } i \text{ to node } j \text{ with weight } w_{ij}.$$

This can be interpreted in various ways depending on the context, such as the number of interactions, the intensity of relationships, or the capacity of links.

It can be seen as a representation of a **multigraph** with N multi-edges: $w_{ij} = N$. If all weights are positive, we have useful theorems for matrices with positive values (Perron-Frobenius theorem) and we can connect to "physical" situations (transition probabilities of diffusion processes). Negative weights can be meaningful too: in **signed networks** the sign of the weight can represent different types of relationships or interactions, such as:

- **Friend-enemy**: in social networks, positive weights can represent friendships while negative weights can represent enmity.
- **Attraction-repulsion**: in physical systems, positive weights can represent attractive forces while negative weights can represent repulsive forces.
- **Ferromagnetic-antiferromagnetic**: in spin systems, positive weights can represent ferromagnetic interactions while negative weights can represent antiferromagnetic interactions.

One can also operate a weighted matrix symmetrization:

- Average symmetrization: $A' = \frac{1}{2}(A + A^T)$.
- Random matrix symmetrization: $A' = \frac{1}{\sqrt{2}}(A + A^T)$.

1.2.3 | Incidence Matrix

We can also represent a graph using an incidence matrix, which is an alternative to the adjacency matrix. The incidence matrix is particularly useful for certain types of analyses, such as flow problems and network dynamics.

Definition 1.3 (Incidence Matrix). The incidence matrix I of a graph $G(V, L)$ is an $L \times V$ matrix (number of links by number of vertexes). It contains -1 for outgoing connections and $+1$ for ingoing connections:

$$\text{if } A_{ij} \neq 0, \quad I_{Ni} = \mp 1, \quad I_{Nj} = \pm 1.$$

It can be interpreted as a "generalized derivative" on the graph, giving the oriented difference between values on adjacent nodes.

Thus the incidence matrix can be used to analyze the flow of information or resources through the network, and it is particularly useful for studying directed graphs and their properties. The incidence matrix can also be used to derive the adjacency matrix through the relation $A = I^T I - D$, where D is a diagonal matrix that accounts for the degree of each node.

Generalized Derivative

The incidence matrix can be interpreted as a generalized derivative on the graph, giving the oriented difference between values on adjacent nodes. The incidence matrix allows us to compute the net flow into or out of each node, which is essential for understanding the dynamics of processes.

Given a vector of values $\mathbf{f}_i$ for each node, the incidence matrix I applied on the vector gives:

$$\sum_j I_{ij} \cdot \mathbf{f}_j = \Delta_i,$$

$$\Delta_i = \mathbf{f}_i - \mathbf{f}_j \forall \text{adjacent nodes},$$

where Δ_i represents the net flow into or out of node i based on the values of its neighbors. This interpretation is crucial for analyzing diffusion processes, random walks, and other dynamic phenomena on networks.

Geometrically, a network acts as a spatial structure: while a regular lattice discretizes Euclidean space, a complex graph represents a more intricate manifold. However, one must be careful, as the properties of this discrete manifold do not necessarily hold in the continuous limit.

1.2.4 | Sparse Representations

For sparse graphs where the number of links is much smaller than V^2 , other representations are preferred:

- **Edge Matrix:** An $L \times 2$ matrix that stores single links instead of the full matrix.
- **Adjacency List:** Each line describes a node and contains the indices of its arrival nodes. This is particularly useful for diffusion processes, such as simulating random walks on a graph.

1.3 | Graph Isomorphism and Permutation

Definition 1.4 (Graph Isomorphism). An isomorphism is a 1-1 correspondence between vertexes that preserves the link structure. It corresponds to a node index permutation.

Remark (Re-indexing). *Node permutation represents node reordering. Re-indexing can be based on a priori information (like intrinsic metrics) or network-based centrality measures, and is very important for network visualization.*

Example (Protein and DNA Contact Maps). A network representation of peptide bonds serves as a discretization of a 3D distance matrix. Spatial closeness acts as an “intrinsic” metric for the graph.

1.4 | Connectedness

1.4.1 | Undirected Graphs

An undirected graph is connected if it is not divided into two or more non-communicating parts. Algebraically, this means there is no permutation that reduces the adjacency matrix into diagonal blocks.

1.4.2 | Directed Graphs

In a directed graph, not every node is necessarily reachable from any other. Connectedness is classified into three types:

- **Strongly connected:** Every node is connected to another by a path, implying the existence of cycles.
- **Unilaterally connected:** For every two nodes, there is at least one path.
- **Weakly connected:** Every node is connected to another by at least a semipath.

Remark (Block Matrix Reduction). *For directed graphs, the adjacency matrix can sometimes be reduced to blocks where nodes in block C are reachable from block A , but not vice versa. In such cases, block C acts like a “sink” or “absorbing state” in transition processes.*

1.5 | Distances and Shortest Paths

1.5.1 | Node Distance

The distance between two nodes is defined as the shortest path between them. In weighted networks, nodes that are “closer” have a greater weight, and distance can be calculated using harmonic composition (similar to capacitors in series). The larger the weight, the smaller the distance.

1.5.2 | Shortest Path Algorithms

The distance between two nodes equals the minimum number of links (geodesic) between them. The diameter of a graph is the maximum path length, while the average path length is the average over all paths. These are typically calculated using Dijkstra’s algorithm (one-to-all) or the Floyd-Warshall algorithm (all-to-all).

1.6 | Subgraphs and Special Network Topologies

1.6.1 | Subgraphs and Cliques

Definition 1.5 (Induced Subgraph e K-Clique). • **Subgraph:** A subset of a graph's nodes and links.

- **Induced Subgraph:** A subset of the nodes of G with ALL the links of G that exist between those specific nodes.
- **K-Clique:** A fully connected subgraph with degree K .

1.6.2 | Complete Graphs

A completely connected graph K_N has all possible links, meaning every node has a degree of $K = N - 1$. The graph complement has the same nodes but the “opposite” links.

1.6.3 | Bipartite Graphs

Definition 1.6 (Bipartite Graph). A graph that connects two different types of nodes between each other, but NOT between themselves. It is represented by an $M \times N$ rectangular adjacency matrix. In terms of matrix algebra, a bipartite graph is an imprimitive matrix of order 2.

Remark (Tensor Contraction). *Given a rectangular adjacency matrix B representing a bipartite graph, we can obtain two square networks connecting nodes of the same type via contraction: $B \cdot B^T$ and $B^T \cdot B$. Nodes of the same type are connected if there is at least one path between them through a node of the other type.*

Example (The Human Disease Network). An example of a bipartite graph is the “disease” (e.g., OMIM database), a gene-pathology network. By contracting the matrix, we can obtain a gene-gene network based on shared diseases, or a pathology-pathology network based on shared genes.

1.6.4 | Trees, Stars, and DAGs

Remark (DAG Adjacency Matrix).

1.7 | Planarity and Homomorphism

1.7.1 | Planarity and Genus

1.7.2 | Graph Homomorphism

1.8 | Network Reduction and Filtering

1.8.1 | Minimum Spanning Tree (MST)

Definition 1.7 (Min/Max Spanning Tree).

1.8.2 | Link Selection Rules

1.9 | Spectral Analysis of Networks

1.9.1 | The Eigenvalue Spectrum

1.9.2 | Powers of the Adjacency Matrix

1.9.3 | Spectra of Specific Graphs

1.9.4 | Isomorphism and Co-spectrality

Remark (Co-spectrality).

1.9.5 | Graph Distances

2 | Random Graph Ensembles

In this chapter, we will explore the concept of random graph ensembles, which are fundamental in understanding the properties and behaviors of complex networks. We will start with the simplest model, the Erdős-Rényi random graph, and then move on to more complex models such as the Stochastic Block Model (SBM). We will also discuss the thermodynamic limit, degree distribution, phase transitions, and other properties of these random graph ensembles.

2.1 | Erdős-Rényi Random Graphs

We are going to start with the simplest model of random graphs, which is the Erdős-Rényi (E-R) model. This model serves as a fundamental null model for understanding the properties of complex networks.

Definition 2.1 (Erdős-Rényi Random Graphs). Networks whose interactions (links) between nodes are chosen randomly, without a priori correlations or rules. In statistical mechanics terms, it is comparable to a “perfect gas” with molecules (nodes) experiencing random “sparse” impacts (links).

It can be seen as a **null model** for complex networks, as it represents a system with no structure or correlations.

If we expect to find a certain property in a real network, we can compare it to the expected value of that property in an E-R random graph. If the observed value significantly deviates from the expected value, we can infer that there is some underlying structure or correlation in the real network that is not captured by the E-R model. It can also be used to confront different systems, such as comparing the structure of a social network to that of a biological network, to identify unique features and commonalities.

If we make the statistical analogy with a gas, the E-R model represents a perfect gas, where the nodes are like molecules that interact randomly without any specific rules or correlations. The edges, which are generated randomly, represent some average quantities of the system, not the distance between nodes or the strength of interactions. The E-R model is a useful starting point for understanding the properties of complex networks, but it is important to note that real-world networks often exhibit more complex structures and correlations that are not captured by this simple model.

2.2 | Statistical Ensembles of Networks

2.2.1 | Microcanonical Ensemble $G_{N,M}$

Definition 2.2 (Microcanonical Ensemble). An ensemble of networks with N nodes, representing a fixed system size, and M links, representing the system energy. The links are randomly selected from the possible $N(N-1)/2$ for a symmetric network or $N(N-1)$ for a direct graph.

Remark (Construction Rule). *The construction rule of one instance of the ensemble is:*

1. Take N nodes.
2. Repeat M times: choose 2 nodes randomly and connect them, checking for loops or existing links.

The probability of generating “bad” links, such as loops and multilinks, is about $1/N$. The effect of this selection becomes negligible for large N .

2.2.2 | Canonical Ensemble $G_{N,P}$

Definition 2.3 (Canonical Ensemble). An ensemble of networks with N nodes (fixed system size) and a fixed probability P to have a link between them, representing a fixed average energy or temperature.

Remark (Binomial Distribution and Construction). *The probability of having a graph with m links and N nodes is governed by the binomial distribution. To construct an undirected graph:*

- Take $N(N-1)/2$ values A_{ij} uniformly distributed between 0 and 1.
- If $A_{ij} > 1 - P$ then $A_{ij} = 1$, else $A_{ij} = 0$.
- Apply symmetrization using logic “OR”: $A \vee A^T$.

2.3 | Thermodynamic Limit and Degree Distribution

2.3.1 | Equivalence of Ensembles

For $N \gg 1$, the canonical link distribution tends to a Dirac's delta function. All instances of the canonical ensemble tend to have a number of links equal to the average link number, thus recovering the microcanonical property.

2.3.2 | Degree Probability Density Function (pdf)

The probability of having a node with k links is the probability of randomly choosing k links among $N - 1$:

$$p(k) = \binom{N-1}{k} P^k (1-P)^{N-1-k}$$

Remark (Bernoulli to Poisson to Delta). *In the thermodynamic limit ($N \rightarrow \infty$, $P \rightarrow 0$) with $P \cdot N = \lambda$ kept constant, the distribution becomes a Poisson distribution:*

$$p(k) \approx \frac{\lambda^k}{k!} e^{-\lambda}$$

In this limit, a random Bernoulli graph tends to a regular graph with equal connectivity for all nodes, even if node proximity is not regular as in a lattice. The Erdős-Rényi model tends to an infinite-dimensional regular lattice.

2.4 | Phase Transitions and the Giant Component

2.4.1 | Average Connectivity

In E-R networks, the threshold for transitions depends on the network size N and the connection probability p . We use the average connectivity λ as the control parameter:

$$\langle k \rangle \approx pN = \lambda$$

2.4.2 | The Giant Component

Definition 2.4 (Giant Component Emergence). • For $\lambda < 1$, there is no giant component.

- For $\lambda \geq 1$, there is a giant component that, for increasing λ , tends to include all the nodes.

Adding one link at a time creates larger and larger clusters, which is known as the “avalanche phenomenon”.

Remark (Transcendental Equation). *If we define u as the probability of a node not belonging to the Giant Component, the average field equation yields the transcendental equation:*

$$u = e^{\lambda(u-1)}$$

The order parameter S , which is the fraction of nodes belonging to the giant component, is $S = 1 - u$.

2.5 | Network Properties in Random Graphs

2.5.1 | Diameter and Average Path Length

Assuming a regular graph ($N \gg 1$) and a sparse network (regular tree, no loops), the diameter scales as:

$$d \propto \ln N$$

A similar result holds for the average path length $\langle l(N) \rangle \propto \ln N$. As the network size grows, the distance between nodes grows slowly, which defines the “small-world” property characterized by easy traversability.

2.5.2 | Degree Correlations

Remark (Absence of Correlations). *There is an absence of correlations between node features. Specifically, there is ALMOST none for finite-size graphs and none in the thermodynamic limit. For undirected graphs:*

$$p(k_1, k_2) \approx p(k_1) \cdot p(k_2)$$

Similarly, in an ER directed network, there is no correlation between in-degree and out-degree: $p(k_{IN}, k_{OUT}) = p(k_{IN}) \cdot p(k_{OUT})$.

2.6 | Stochastic Block Model (SBM)

Definition 2.5 (Stochastic Block Model). A more complex null model that can modelize a non-trivial community structure of the network. It can be obtained by combining diagonal E-R blocks, possibly of different sizes and link densities, with non-diagonal blocks of random links.

Appendices

A | Notation and Conventions

B | Computations and further developments